Data-efficient Large Vision Models through Sequential Autoregression

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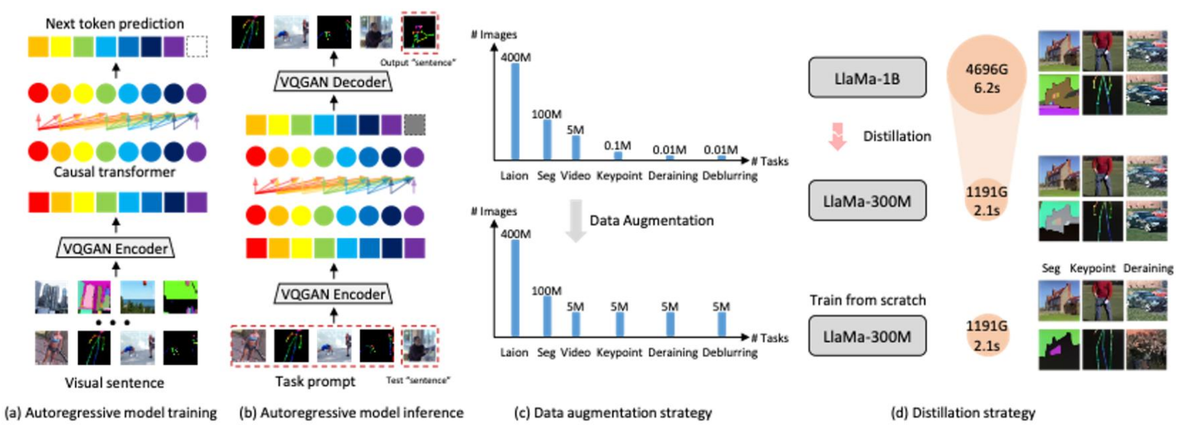
PyTorch code

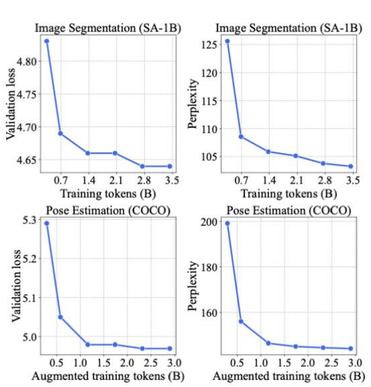
Introduction

Training general-purpose large multimodal foundation models on purely sequential data, similar to linguistic inputs, has heralded a new frontier in visual perception. However, current endeavors are hamstrung by an over-reliance on colossal models, exemplified by the necessity for an extensive corpus of data, often comprising staggering 400B tokens. In this work, we delve into the development of an efficient, autoregression-based foundation model, innovatively architected to operate on a limited dataset. Our evaluations demonstrate how this model achieves proficiency in a spectrum of visual tasks spanning both high-level and low-level semantic understanding. With a significant reduction in parameter footprint, and a marked decrease in training data requirements, we pave the way for more sustainable and accessible advancements in the field of generalist visual models.

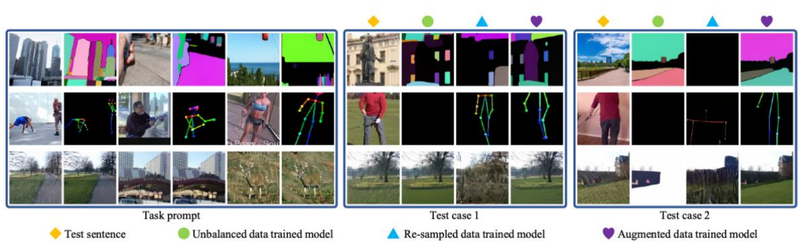


(a) Autoregressive model training (b) Autoregressive model inference (c) Data augmentation strategy (d) Distillation strategy

Part-1: Data-efficient Training: Data Augmentation Strategy



Single task



Dataset configuration	Image Segmentation (10% of SA-1B: 3.434B tokens) loss ↓	accuracy ↑	perplexity ↓	Pose Estimation (COCO-Pose: 0.029B tokens) loss ↓	accuracy ↑	perplexity ↓	Image Deraining (Rain13K: 0.007B tokens) loss ↓	accuracy ↑	perplexity ↓
Unbalanced	4.48	20.32	88.59	4.95	19.86	141.91	5.69	11.89	279.88
Balanced by re-sampling	5.01	16.68	151.66	16.11	15.75	-	21.73	4.53	-
Balanced by augmentation	4.68	18.79	107.77	4.94	20.24	140.39	5.64	11.57	269.04

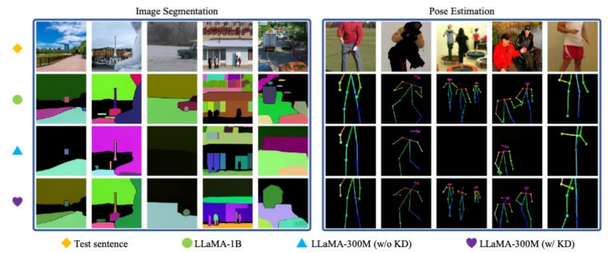
Multiple tasks. "-" indicates collapsed results.

Dataset augmentation yielded the best overall performance.

Part-2: Low Power Consumption & Tiny Model: Knowledge Distillation

The introduction of KD proves beneficial in enhancing the single-task performance of LVMs.

Model	KD	loss ↓	accuracy ↑	perplexity ↓
Image Segmentation (10% of SA-1B)				
LLaMA-1B	-	4.50	20.18	90.04
LLaMA-300M	✗	4.64	19.17	103.24
LLaMA-300M	✓	4.59	19.48	98.72
Pose Estimation (COCO-Pose)				
LLaMA-1B	-	4.90	20.96	134.07
LLaMA-300M	✗	4.97	20.46	144.08
LLaMA-300M	✓	4.91	20.80	135.91



The introduction of KD proves beneficial in enhancing the multi-task performance of LVMs.

Model	KD	Image Segmentation loss ↓	accuracy ↑	perplexity ↓	Pose Estimation loss ↓	accuracy ↑	perplexity ↓	Image Deraining loss ↓	accuracy ↑	perplexity ↓
LLaMA-1B	-	4.55	19.72	94.75	4.86	20.77	129.95	5.53	12.20	245.33
LLaMA-300M	✗	4.68	18.79	107.76	4.94	20.24	140.39	5.63	11.57	271.38
LLaMA-300M	✓	4.67	18.84	106.81	4.93	20.32	139.27	5.62	11.93	269.04

LLaMA-300M w/o KD vs. w/ KD on LSP dataset.

LLaMA-300M w/o KD vs. w/ KD on OCHuman dataset.

Query Image, LLaMA-300M w/o KD, LLaMA-300M w/ KD, Ground Truth on the rain100H de-raining dataset.

Beyond Perception: Generation Ability

Next frame prediction: zoom in, zoom out, camera motion.

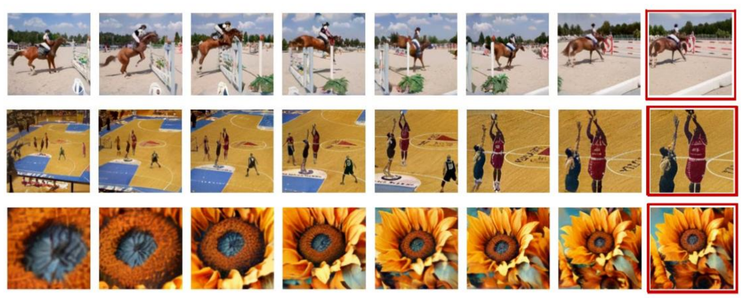
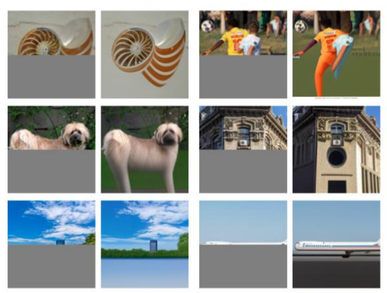



Image completion.